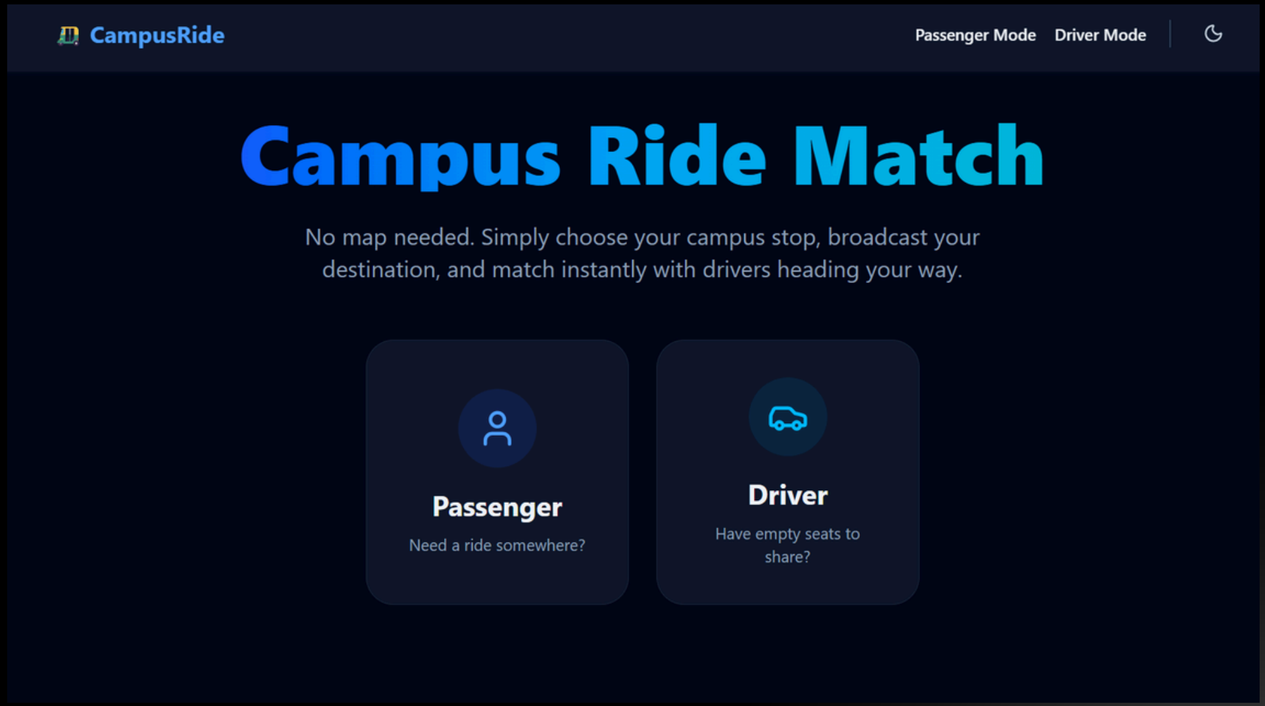
CAMPUS RIDE

Map-Free Rickshaw Matching System

Group 20
- Milan Prajapati
- Soumith
- Sujith
- Matturty Sri Lakshmi

CAMPUS RIDE-SHARING SOLUTION



PROBLEM & SOLUTION

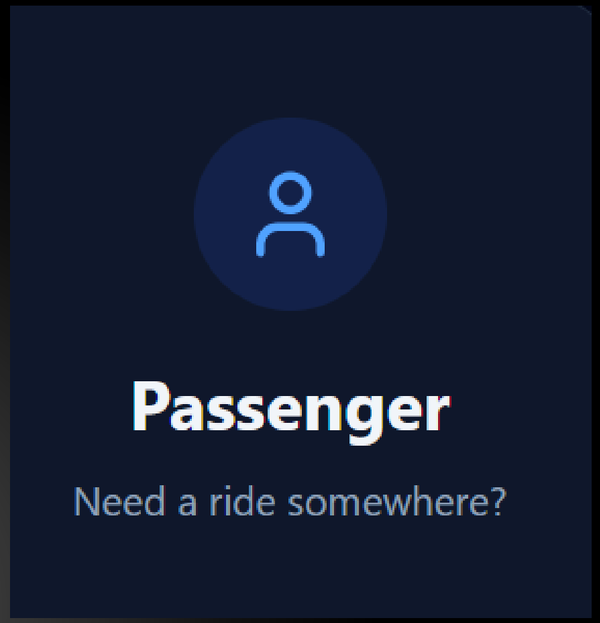
The Problem

- Standard ride-sharing apps rely on GPS maps not available inside campuses
- Campus routes are based on informal stops (Library, Hostel, Gate) rather than mapped roads
- No centralized system to connect passengers with rickshaw drivers
- Passengers spend time manually searching for available rides

Our Solution

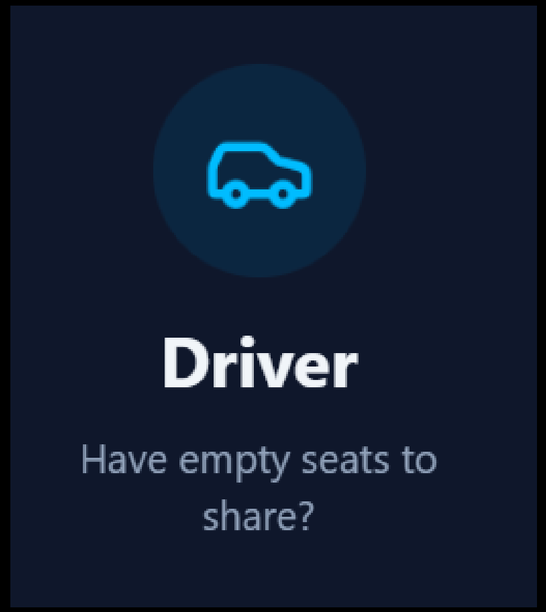
- Map-free ride matching using predefined campus stops
- Passengers select pickup and destination from a verified stop list
- Ride requests broadcast instantly to nearby drivers
- Drivers accept requests and manage available seats
- Real-time notifications with driver details and estimated arrival

FEATURES



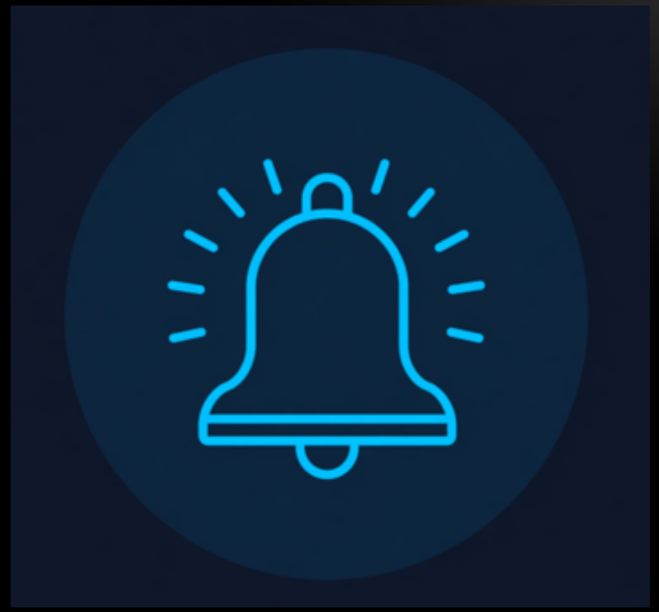
Passenger Features

Select stops, broadcast requests, receive notifications, view driver details & ETA



Driver Features

Receive broadcasts, view stops, accept rides, automatic seat management

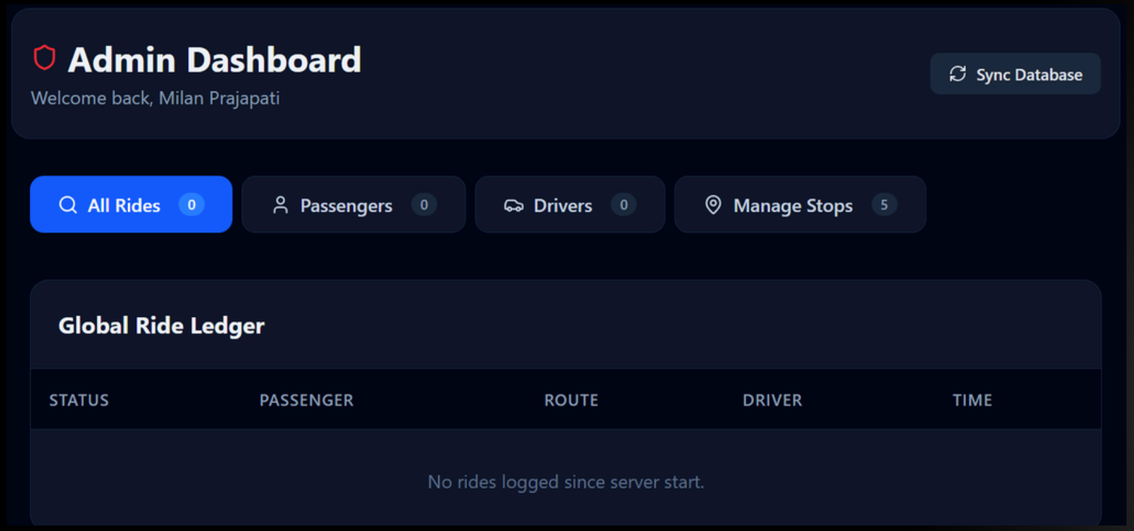


Real-Time Updates

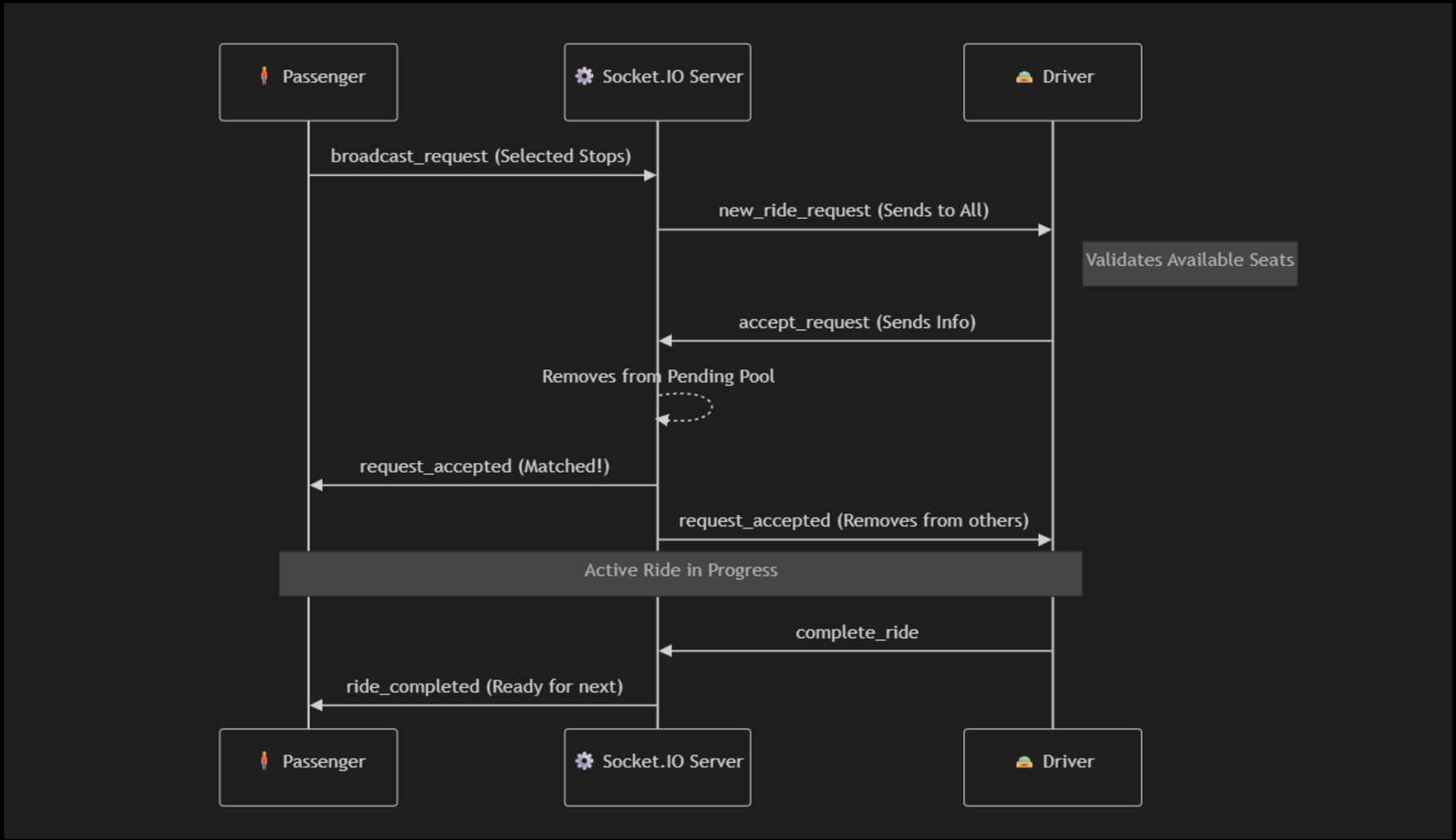
Instant notifications, no refresh needed, simple authentication for all users

ADMIN FEATURES

- Add, edit, delete campus hotspots
- Track ride logs with timestamps
- Monitor connected passengers and drivers
- View ride booking and completion status
- Manage system data through dashboard
- Control stop list without changing code



WORK FLOW


Passenger
Socket.IO Server
Driver
broadcast_request (Selected Stops)
new_ride_request (Sends to All)
Validates Available Seats
accept_request (Sends Info)
Removes from Pending Pool
request_accepted (Matched!)
request_accepted (Removes from others)
Active Ride in Progress
complete_ride
ride_completed (Ready for next)
Passenger
Socket.IO Server
Driver

THE TECH STACK

1. **Frontend:** Javascript, Typescript, React
2. **Styling:** Tailwind CSS
3. **Backend:** Node.js, Express,js
4. **Real time Communication:** Socket.io
5. **Database:** Firebase

TEST CASES & EDGE CASES

- Same pickup & drop-off stop blocked; empty fields disable submit
- Page refresh restores full ride session via sessionStorage
- Driver's Accept button disabled when seats are full; duplicate requests filtered
- Accepted ride instantly removed from all other drivers' lists
- Wrong admin credentials block all access; delete stop requires confirmation
- Empty stop name blocked before sending to server
- Only the correct passenger receives the driver match notification
- New drivers receive all pending requests on connect; listeners cleaned up on unmount

TEAM CONTRIBUTION

- Milan Prajapati - Backend Architecture, Admin Portal, Socket Matching
- Soumith - Passenger & Driver UI, Frontend Socket Integration
- Sujith - Backend Initialization, Firebase & Server Structure
- Matturty Sri Lakshmi - Frontend Setup, Home Page, Dark Mode & Styling

CONCLUSION AND FUTURE WORK

Conclusion

- Introduces a map-free ride matching system for campus use
- Reduces waiting time and improves ride availability
- Enable real-time communication between passengers and drivers
- Efficient utilization of seats and resources
- Simple, scalable, and easy-to-use system

Future Work

- Develop a mobile application (Android/iOS)
- Add online payment integration
- Introduce GPS support as optional enhancement
- Improve notifications with SMS/alerts